

Secure Login System Workflow

Below are the detailed workflows representing the different defensive layers and logical flows built into your Secure Vault application.

1. User Registration Workflow

This flow ensures that no weak passwords are created and that all sensitive data is securely hashed before touching the database.

[image]

2. Authentication & 2FA Workflow

This is the core login sequence. It evaluates rate limits, locked states, validates the hash, and conditionally requests a TOTP token if the user has 2FA enabled.

[image]

3. 2FA Setup Workflow (Post-Login)

When a user decides to turn their account into an impenetrable fortress by adding 2FA.

[image]

4. Threat Monitoring & Dashboard Workflow

Once inside, the application maintains security visibility based on your **Role-Based Access Control (RBAC)**.

[image]

What Happens in the Database During a Login

If we peeked into the `activity_log` table during an average user's session, it looks like this:

Timestamp	Action	IP Address	Status	Notes
10:05:12	Registration	192.168.1.1	Success	Admin account created
10:06:01	Login	192.168.1.1	Failed	Wrong password
10:06:22	Login	192.168.1.1	Success	Correct password
10:08:45	Setup 2FA	192.168.1.1	Success	Scanned QR successfully
11:30:00	Login (2FA)	192.168.1.1	Success	Passed both Password + TOTP
11:45:10	Logout	192.168.1.1	Success	Session cleanly destroyed